

ActInf Textbook Group ~ Cohort 3 ~ Meeting 17 (Chapter 7, part 2)

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Hi, thanks everyone.

It's 4th of July, 2023, and we're in our second discussion on Chapter 7.

So we'll head back over there.

Does anyone want to type a question or add any reflection or thought or just question or area they want to look at today?

Okay.

Well, I could say something.

Hi there.

Yeah, please.

So last time I joined was four weeks ago, and I was saying about while I was advocating having Python code as examples, I went away and just coded up.

The bits at the very beginning of Chapter 7 of the notes, the musical notes, which were sort of transferring, converting the text into Python with these comments.

It didn't really get me.

Well, it showed the reasons why the matrices contain the numbers that they did.

It didn't really say much more than that.

But I've just discovered Appendix C in the book.

And in a way that goes through something similar.

Although it doesn't explain the matrices, the precise numbers in each matrix in quite so explicit a way.

And then later on in Chapter 7, there's the maze example.

And again, these are all filling in with the application.

But then it's just formatting the data to the point where it can then be given to an SPM12 function.

Yeah.

So MDP there without.

So I would like to see what's inside there and then be able to graph, for example, various metrics like equations.

Things like mutual information, novelty, to help get an intuitive feel.

I think Appendix C says about being able to play around with the numbers, get an intuitive understanding.

But I want to see more than just playing around with the matrices, the inputs.

So, yeah.

So that's just an update on where I am, really.

Thank you.

Yeah.

Appendix C is very sparse.

Mm-hmm.

And it doesn't.

I mean, they would have done well, potentially, to have just simply provided the reproducibility code for each figure without interspersing it with plain text.

Mm-hmm.

Because then it's kind of this reverse engineering process.

But what you said is absolutely right on, which is the entire POMDP, the entire figure 4.3 discrete time model.

It's all the letters that we talk about.

A, B, C, D, and then E, which is just given the letter V here.

And then the true starting condition.

And then you just let it run.

Now, here's the SPM, MDP, VB, X, variational base, X.

I'm not sure exactly what the X is.

But I've heard that this is sort of a, like, not inscrutable, but it's not simple to translate this function just straightforwardly to Python,

which is one reason why, like, many people do continue using the MATLAB variant, because this contains a lot of, like, accelerated methods that are -

and it contains, like, a lot of extra bells and whistles to generate the figures, and so that makes it quite easy.

But in general, it speaks to exactly what Chapter 6 highlights, which is that active inference modeling is about representing the agent and the environment,

and then the methods for letting the agent come to its own solutions are straightforward.

Rather than needing to do a secondary - and at some computational scale, yes, further techniques are needed.

But to a first approximation, the work of understanding a system of interest is in exactly specifying the dimensionality of the matrices.

Same as in pi MDP in model stream 7.

But yeah, how was it - I mean, did you see the matrices any differently or think about the models any differently?

No, what I coded up, it was really just - well, to get an idea of how things might be coded up, and this wasn't to have something that -

certainly not anything that was simulation efficient, it was something very clumsy.

So, for example, what you're showing there with the B matrix, it would be explaining why the 97s are there in those particular places.

So, starting off with an all zeros matrix, or maybe an all ones in this case.

And I know it's normalized there.

But so that you can - you're then describing the matrix B directly in terms of the application.

You don't need the separate bits of text to explain what there is there.

So I think that was helpful, or would be helpful as part of the wider if that was taken to completion, and possibly the team A's one could be taken further, and would have been more useful.

Yeah.

But yeah, for me, it's getting inside those functions to see what's going on.

I don't have any real feel.

So I'd like to be able to see, well, here's free energy changing as various things happen.

And it's comprised of these different factors, or it can be interpreted in this different way,

so that you can then play around with the numbers, say, to put it into a position where there is novelty, and you can see the aspect, the term of free energy that's described as novelty, and see the effect on that.

So I think that would give me a much better feeling rather than the SPM 12 is just a black box solution.

Yeah.

Yeah.

Yeah.

Cool.

And if you want to, we can host on the textbook group repo.

So maybe we can then use it and, you know, chapter seven, eventually have every single example, potentially even in multiple languages, but sorted like according to the numbering of the textbook.

Yeah.

Yeah.

So at the end of probably cohort two, I, there was, I put some comments of, you know, I'd be interested in setting or being involved in a group that was basically doing what I've said.

Well, to, to, to, to, to, to, to, to pick, um, by MDP and someone made a comment of, well, you could just code up the equations, um, that are in the book and maybe, maybe that's the right approach.

Um, cause I've, I've looked into SPM 12 and the data structures in there are just horrendously complicated, really.

um so so trying to get into that would be you know would be very difficult um but

maybe just building up from equations in the book would be uh would be manageable

cool

cool thank you

um anyone else want to bring up anything or ask anything or we can look at the questions

okay not I don't remember exactly which ones we may have addressed um last time but

um okay okay that's

what's happening here

logged in under a difference

okay

those are the thumbs up that I knew had to be there okay um

um

what do the rows columns numbers mean in equation 7.2

to check your understanding could you design another b matrix that could be used in the music

example or another example so the simple answer is each cell represents the probability of moving from a row to a column and these are the notes one two three four with this one over a hundred so that

summing across you get um one

um this musician it has a 97 percent accuracy um uh correctness in in moving forward through this song and uh as we look towards what kinds of like questions and variants would help us um deepen understanding like

what would it look like to be more accurate well there'd be a 98 or what would it look like if the third note to the fourth note was really hard or what would it look like if there was five notes

or um write a 12 tone composition using 12 notes but anyone else want to have any comments or thoughts on this

okay

pretty

straightforward

question

okay

in the caption of figure 7.2

it says lower left negative free energy gradients i.e prediction errors through time

well

first issue it's empty

that may have been fixed in in some subsequent like not a full reprinting but that may have been fixed but it's empty there

what are similarities or differences between prediction error and free energy gradients are they the same thing what's the difference between free energy and free energy gradients what is a gradient um

here i'm going to look to actually

this question which we looked at earlier today

so this question from chapter two asked

um

prediction error surprise and variational free energy

they're related to each other but it's sometimes confusing

and um

to to kind of look at this again

the best fitting model

with the most model evidence

is going to have the lowest prediction error

the lowest lowest surprise and the lowest variational free energy but there are subtle differences

prediction error

is a simple

differential

between

prediction
and
observation
so if
the prediction is 10
and the observation is 11
the prediction error
is
positive one
it's a
it's in the units
of whatever the observations are about
temperature
surprises and information
theoretic
quantity
every single
um
data point
observation
is on a distribution
is associated with some
non-zero
surprise value
but it can be minimized
and it's minimized when the prediction error is lowest
and variational free energy
is described
in
equation 2.5
so it's not exactly the same
as surprise
surprise but
it's a bound on surprise
that's kind of the whole point
and
it's lowest
when all of these things are kind of

in unison
and then
Ali also highlighted that
surprise is what is calculated
in exact Bayesian inference
where model evidence is maximized
via surprise minimization
however
where exact Bayesian inference
is intractable
then

we utilize
approximate
Bayesian inference
and
there's two ways
that you could go about doing that
one approach would be
a sampling based approach
but that would be enormously
computationally costly
in certain settings
and you wouldn't really know
when you've reached
the right
time to stop sampling
but
variational inference
allows you
to
um
approximate
Bayesian inference
by bounding the surprise
with a free energy
and that's the evidence
lower bound

and it gets to the last piece
of the question
which is
what is a gradient
so a gradient
is a rate of change
it could be
um
on a two-dimensional
um
surface
the rate of change
like just a ruler
at the angle
of the rate of change
of that local point
you could have a gradient
in one dimension
you could have
um
um
different dimensionalities
of gradients
but basically
they're the local rate
of change
around some point
and so it's like
if you have beliefs
about
um
the
temperature
and they're 10
and then
the observation
comes in
and it's 11

so that's associated
with some prediction error
some surprise
and some free energy gradient
and then
the gradient
should steer you
towards updating
your prediction
from 10
upwards
under the full attention
setting
you would go from
10 all the way to 11
under a no attention
setting
you wouldn't move at all
but you'd want to move
from 10
towards 11
that's the gradient
or the flow
going towards 11
whereas you wouldn't
want to go uphill
towards like 9
thank you
Ali
cool
let's replace this
in the figure
I'll do it
okay
any other thoughts
on on this
little bit unclear
with the

dark lines
because they're all
crossing and everything
like that
but the kind of
moral of the story
is
when something
happens
there's a transient
adjustment
of the model
but also
as we've
discussed
like
we're in
discrete time
so it's
it's not
immediately clear
why there's
interpolated
continuous
values
okay
maybe this is
speaking to this
yeah
so figure
7-2
and 7-7
have these
curves
where belief
updating is
happening
faster

and the
text alludes
to some
sort of
simulation
machine
yeah
I believe
that is
the
stock
machinery
of the
SPM
package
but I'm
not
exactly
sure
yes
it does
include
like
when
just like
when we
looked at
the
SPM
VBX
it has
a section
on
dopamine
and I
think that
that is
being

displayed
in
continuous
time
interpolated
but I'm
not exactly
sure why
that happens
and I think
it would be
more
clear
given
our
discrete
time
setting
to just
show a
point
associated
at each
time
step
does anyone
have
any other
thought
on this
Thank you.
Thank you.
Thank you.
Okay.
Let's look at what others have.
Okay.
Okay.
Okay.

Okay.

So you can have the joint distribution of let's just say A and B here, but we're not talking just only about the A and the B matrix, but actually it turns out that those are factorizable.

So it allows you to break down the statistical joint estimation of A comma B into A times B.

If they did influence each other, then it wouldn't be factorizable.

So neurologically, one of the most common places where people discuss factorization is with the what and the where stream in the brain.

And the idea here is like an object's identity should be factorizable from its, for example, location in the visual field or size visually.

So you want to be able to separate out what something is and where it is.

And people have identified these, these two streams of visual information flow that reflect the, the, the, the, the what and the where.

So I'll bring this in, but the what and the where are these two.

Oh, funny.

Pazoa.

These two streams of visual input.

So people have, have kind of like, it's, it's one of those things where it's like the structure of the inference architecture recapitulates the structure of the problem.

Because again, we, we would expect vision to have that kind of factorization between what an object is and where it is.

And here.

Someone has added an example about fruits and berries, different properties.

But they're all kind of related to ripeness.

And from that from that from that from that from that from that from that from that from that from that from that
from that from that from that from that from that from that from that from that from that from that from that from that
from that from that from that from that from that from that from that from that from that from that from that from that
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from that from that from that from that from that from that from that from that from that from that from that from that
from that from that from that from that from that from that from that from that from that from that from that from that
function. So, you know, the words, the factorization system for a given category consists of two,
two sets of, or two, two sets of morphisms. So one of which have the ability to, to contain all the

isomorphisms of the given category, and also they're closed under the composite function. So, you know, in this sense, factorization system can give rise to the concepts, some other concepts in category theory, such as, and also an important property of factorization system is each factorization is itself a functorial. So, two, I mean, they can be regarded as two morphisms that can be mapped onto each other by commuting arrows or morphisms. So, yeah, it's one of the important concepts in category theory, as it gives rise to some other concepts, such as equivalency and orthogonality and so on. Cool. Thank you.

One other point, and the reason why factor graphs and factorizations are even being discussed is the the connections in the Bayesian graph reflect which variables have influence on each other.

So, for example, O of T minus one is only influenced by hidden state of T minus one.

So it's like through the A matrix and similarly. And so it's not the case that all variables have an edge to each other. First off, that would cause connections like through time, which would violate the Markovian property, which says that the only way that the past can influence the future is through the present.

So that's a massively simplifying assumption. If you think about a time series with 100 time steps, if you enable every time step to be possibly connected to every other time step,

you're going to end up with calculated output, however many hundreds and hundreds of edges.

Whereas if you abide by the Markovian property, which is at least a defensible statistical approximation, if not even more valid than that, the Markovian property in time simplifies out how many edges you have to fit.

But also there are sparseness about like π doesn't reach down to influence observations.

And so that, first off, captures something important and structural about the problem we're trying to solve.

We don't want to even open the door to an opportunity for policy to just be like changing our observations with a thumb on the scale, nor do we want policy to simply intervene directly in a hidden state.

But rather with the B matrix, we have policy intervening in how hidden states change through time.

So the sparseness relative to the all by all edges that could exist,

the sparseness helps embody the statistical and causal nature of the problem.

And it's massively computationally simplifying.

So that's why factorization comes into play.

Related factorization and the recipe of chapter six.

Why are there two A matrices?

So let's just clarify which example.

So this is probably about the maze.

Yeah.

Okay.

So this is about the maze on page 131.

Okay.

Okay.

Didn't see that.

There are two modalities that represent exteroceptive sensory data pertaining to where.

So that's just the location of the rat in the maze.

And also there's a what modality.

That is like, am I tasting something delicious or not?

And so just like the what and the where in the brain,
there are two matrices and they have different shapes.

And so that's what I'm saying.

Because the location, the where A matrix maps location cues to location hidden states,
true locations, which is here a noiseless mapping.

And the second modality maps location to whether there is no tasty queue or aversive queue at these two locations.

So when the mouse is in the center junction or down in the bottom, we're in this dashed line row.

And then in this example, the right side has the black circle and the left side has the white circle.

Any other thoughts or comments on this?

Okay.

I think this is a little test question.

Don't know if we left this for ourselves or someone else is dropping test questions for us.

Walk through the section from 131 to 135.

Explain all the rows, columns, subscripts, and values.

We just described the A1 and A2 matrices.

Columns are locations.

In the A1 matrix, these are queues of locations.

And then in the A2 matrix, these are what is at the location.

So where and what.

This is the same, except now the black is on the left instead of on the right.

And so we see that reflected with here.

We have a 1.

And a 98 on the right.

And a 1 on the bottom.

Here, the 1 has moved up a row.

And the 98 is on the left.

B2 is not influenced by policy.

1 stays at 1.

And 1 stays at 1.

So it's kind of like A matrix having the diagonal.

The identity matrix for A is kind of like the trivial, noiseless perception.

Observations just map one-to-one to hidden states.

And similarly, a B matrix with an identity diagonal is kind of like trivial action.

Because if you're in this state, you stay there 100% of the time.

And if you're in that state, you stay there 100% of the time.

But that's B2.

They're just getting that out of the way.

They're basically saying whatever you do, you're not changing the structure of the problem in terms of the location of the food.

And that's saying that the rat has no control over the context.

However, they are going to focus in on the four slices of the B matrix.

So the B matrix is intermediating hidden states and how they change through time.

So the hidden state S is about the true location that the rat is in.

So there's four locations.

And the B matrix is going to describe the transitions in location under the four different action policies.

And so there are certain states that you can't.

Well, this is four matrices that the rat can choose.

They allow for a move from any state except for the left and right arm to any other state.

However, the right and the left arms are absorbing states.

The rat must stay.

So here we have the absorbing state.

Here's one.

If you're on the left, you stay on the left.

And if you're on the right, you stay on the right.

And then here, whether you're here or here, you transition to going to the right.

So I guess I can jump from here to there.

And C is the preference vector.

The C1 is for, again, the location.

And this just kind of pushes it off the starting location.

So it does something.

And then here in the C2 is the preference over the context observations, like the preferring the tasty food, not the aversive.

And then just note that because these are in, like, natural log, kind of, or they're an e to the exponentiation.

A six means that the attractive is e to the sixth more rewarding or, you know, more preferable.

D is the prior.

It has a perfect true belief about its starting location.

And regarding the context, it has a 50-50 uncertainty about the equal probability of which context it's in.

But this is a good question.

So there we have the two matrices, the one and two.

So going back to what we were saying earlier, if you were going to feed that into SPM12, would you then multiply those A1 and A2 together as a chronic matrix to get something that was something like maybe 16 by 15 there, a big matrix?

And what we're saying is, in this application, you're able to, the what and the where are separable.

But to use SPM12, you would have to multiply these together.

So a Kronika product or, yeah, is that what would be done?

I think we'd have to look.

But my understanding is that the S true location and the true, like the true awareness would be multiplied by this A matrix to give the where observation.

And then the true context would be multiplied by this, which is uninformative about which context it's in if you're in the bottom here.

And so the true context would be admitted through A_2 to the context observation.

However, there's many situations, speaking a little more generally, where you can hack it into a huge single matrix.

Like you could imagine there's four locations.

Or just speaking of the observations, you could have five locations with null reward, five with the black and five with the white.

And a 15 across or in a 15 row matrix where every combination of location and context is a unique one.

Yeah.

But that defeats the whole point of factorization.

Yeah.

Yeah.

The factorization is usually more simple.

Yeah.

Yeah.

Yeah.

Yeah.

Except there may be some interesting settings where very, very large, sparse matrices may have some special computational representations that actually might make that efficient.

But not in this scale.

How can we read equations such as 7.8?

Okay.

This is a little bit of a general comment that might be better fit for the math learning group.

I'll just throw it over there.

Okay.

And now to 7.8, we don't have the natural language description, human written.

However, let us go to the AI.

See how it does.

Okay.

7.8, LaTeX, Coda AI columns.

Okay, this is a pretty straightforward reading.

The epistemic value I of π is being calculated based upon the posterior predictive entropy and the expected

ambiguity.

Not a very helpful explanation.

This one is why would a high school student be interested in it?

They may not be interested.

Think higher of high school students.

Here's a column asking about where it might be important, and here's defining the variables.

So, not super helpful for exactly what we're looking for, but it gives a look.

But to return to the question of reading it.

Anyone want to just pick a line and read how they see it?

Okay.

Okay.

I won't go verbatim, but the epistemic values.

We're just looking at the epistemic aspect of the equation 2.6 expected free energy.

H is an entropy, so it's describing how blurry or peaked a distribution is.

And this is about the observation distribution through time, $\tilde{o}$, conditioned on action.

So, if our action is associated with very blurry observations, then it's going to be high entropy.

Whereas if the observation entropy is very low, then those observations have lower entropy.

So, if we're getting a broad spectrum of observations from policy, that would have higher epistemic value.

Whereas if you know that every time you open this book, it's going to give you the same 100 time steps, the same data, then it's not as informative.

And here is an entropy about essentially the A matrix.

Because it's discussing the relationship between the observations conditioned upon hidden states.

Also being conditioned upon the policy.

But this kind of goes without saying about the conditioning over policy.

Because basically everything is going to be conditioned upon a policy.

Because the epistemic value is something that's computed on a per policy basis.

Here.

Mutual information, which Neil alluded to earlier.

It's a very interesting quantity.

It's a very interesting quantity.

Especially because it's an equivalence.

We have Q of S given π on both sides.

So, this part is kind of not changing for a given policy.

And here we have P of observations given hidden states.

It's like the A matrix.

Relationship to the variational posterior on observations given policies.

So, again, everything is conditioned on policy.

So, it's kind of like P, O, S .

Difference or divergence with Q of O .

So, if S is...

If the O 's are the same and S doesn't change anything, then the divergence is low.

If you can imagine if S added nothing, conditioning on S added nothing.

You know, conditioning on the true temperature, the sequence of thermometer observations was the same.

Then, knowing the true temperature didn't help you at all.

So, the mutual information would be low.

Whereas, if knowing the hidden states were to provide information, then P of O conditioned on S would differ from Q of O .

I'm not 100% confident on any of these readings.

I'm just kind of just saying them.

And this last one is, again, after you kind of ignore the pies here.

It's Q of S , Q of S conditioned on observations.

So, if the observations are meaningless, then this term is zero.

Whereas, if the observations do matter, conditioning on O does matter, then there is a positive information gain.

And I'm not exactly sure what this little supplement is adding.

Other than describing the factorization relationship, it's saying the variational distribution of hidden states through time, condition on policy and observations.

So, kind of like our inference about true temperature, depending on what we do and observations from the thermometer, is defined as equal sign with a triangle.

And here's essentially a factorization.

Probably derivable from the Bayesian theorem and the sparsity of the generative model.

P of O, S is like the A matrix.

Q of S given π is basically the B matrix.

And then this is the downstairs bit.

Which was seen up here.

Ali?

Thanks.

Yeah, well, if I may give a kind of more informal reading of this equation.

So, here, well, this equation basically means that both, I mean, extrinsic and epistemic value work together to somehow increase the likelihood of our preferred outcomes by minimizing the uncertainty.

So, for example, the extrinsic value depends on the posterior predictive entropy.

Entropy, as shown in the first line, which can only be informative if the agent can be certain about a current state or its current state.

So, it means that epistemic uncertainty should first be resolved before, I mean, any kind of action or any kind of utility will come into play.

And also, at the same time, if the agent is confident enough in its current state, it shouldn't somehow

undertake or perform in a kind of epistemic actions that would seek to pursue a kind of successful plan.

So, and especially this is related to the exploitation, exploration dilemma.

So, briefly, as also we talked about in the previous session, well, minimizing free energy corresponds directly to approximate Bayesian inference by choosing the least surprising outcomes.

So, if the agents could somehow model their environments, they need to have posterior beliefs about the amount of the control they have upon the state transitions producing those outcomes.

In other words, we need to somehow take into consideration the posterior beliefs about those control states, which by turn relies on prior beliefs about those controlled outcomes.

So, yeah, I mean, by using this kind of principled way of resolving the uncertainties, or in other words, resolving the prior uncertainties based on the control states that can minimize the expected free energy, we can somehow achieve a theory that can provide a formal definition for that epistemic value or the value of the information we can gain from the situation.

And also, it somehow focuses the importance of the purposeful behavior that also relies on the generative models that generate those or consider those future outcomes.

So, this is basically the formulation that includes or unifies some of the perspectives or pre-established perspectives, such as the KL control theory and many other previously established perspectives and approaches.

So, in case anyone wants to further explore the derivation of epistemic value and the implications of it, I highly recommend one of the earlier papers by Friston et al., namely Active Inference and Epistemic Value from 2015.

Oh, awesome. Thanks.

Thanks.

It's in 2015.

Great. I'll add it in here.

Yeah.

A lot of, well, we know pragmatic value.

Unlike reward maximization, our active inference concept of pragmatic value is the alignment between preferences and observations.

So that does the kind of kernel work of preference satisfaction and homeostasis.

And then epistemic value, depending on the sophistication of the generative model and what other features it contains, like learning and nested structure, attention, and so on.

Epistemic value does a ton of work.

And that's exactly what we point to as what is missing in, for example, reward learning.

Two huge issues.

First off, it works well when you know what the reward function is, but discovering it is an issue.

And even when you do, and if you do discover it, or a proxy or a heuristic or however you want to think about it,

all you've really done is just coerced, shoehorned different components of epistemic value into the currency of pragmatic reward.

So with equation 2.6 and expected free energy, we get to delineate epistemic and pragmatic value.

And then the challenge of parameterizing these models is to balance them.

For example, if the preference vector of the maze was 6.1 billionths, then maybe it's not as big of a deal in comparison to the epistemic value.

And in contrast, where the preference is 100 million versus negative 100 million, you see more overall pragmatically oriented behavior.

So expectation pursuing, aversion repulsing.

Well, an interesting chapter.

Learning.

More technical details.

Some equations that we haven't always gone through.

But again, I hope for all of these equations that we end up with natural language and pseudocode and code representations.

More examples that are just sort of briefly touched upon.

Massive topics like structure learning just being essentially alluded to.

And hierarchically nested models.

In this case, a discrete time and a discrete time model.

You can tell because of the minus 1 t , t plus 1.

Little figure 4.3.

Fractal figure 4.3.

And then another kind of discrete state space, discrete time.

But here, a nested model.

It's kind of like a 3-4 song.

Like 1, 2, 3, 2, 2, 3.

With this nested timescale.

That's 7.

Well.

Next week, we turn to 8.

Continuous time.

I think that we've hopefully surfaced a lot of opportunities for improvement of the questions and discourse code equations, explanations, and so on.

Because 7 definitely is moving lightly through multiple examples and results and topics.

And so I think there could be a lot of good work with us sweeping our trail and adding questions that we wish would have been asked or would have been added to connect the dots.

And then chapter 8 is going to be a lot of different types of things.

And then chapter 8 is going to head into the continuous time setting and close on hybrid models, which have continuous and discrete time components.

Before chapter 9 goes into the database modeling.

And chapter 10 concludes it.

So, thanks, everyone.

Great times.

Looking forward to, yes, see you at the alternative or whatever time next week on Lucky 7-Eleven.

Okay.

That's great.

Thank you.

Farewell.

Bye.

Bye.

Bye.

Bye.